

Engineering Medical Information Systems: Architecture, Data and Usability & Security

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Abstract

There has been increasing pressure on the health care sector to adopt information technologies to rationalize service delivery and increase service quality. Medical information systems need to be highly interoperable and effectively manage complex information of great sensitivity. Moreover, they have to be optimized for usability in a highly complex knowledge base and agile work environment. This tutorial introduces key concepts, methods and techniques essential for engineering clinical information systems, in particular electronic medical records. It targets participants with basic software engineering knowledge who are or will be involved in development, maintenance, evolution or research of medical software.

1. Summary

Electronic medical information systems (MIS) have been identified as a key enabler to help keeping health care systems sustainable. A 2005 RAND study estimates that the U.S.A. could save \$81 billion annually and help to improve the quality of care through the adoption of high quality MISs. Governments around the world have begun to aggressively promote health information technology. As a result, public and private organizations have an increasing need to recruit software engineers with an understanding of the domain knowledge needed for developing, evolving and maintaining MIS. Moreover, several interesting research challenges exist in this emerging area.

The objective of this full-day tutorial is to respond to this growing need and to offer ICSE del-

egates an introduction to key concepts and techniques required for engineering MIS. In response to the interdisciplinary nature of this subject, the tutorial is jointly presented by a software engineer and a medical practitioner and health informatics researcher. The main topics of focus are:

1. Architecture & Infrastructure
2. Health Data and Information
3. Clinical Knowledge
4. Usability
5. Privacy & Security

2. Architecture & Infrastructure

Different kinds of MIS exist, ranging from clinical point-of-care (POC) systems to public health information systems for disease surveillance and strategic decision-making. We will start by introducing a taxonomy of the MIS ecosystem [1]. Among the various categories of MIS, we mostly concentrate on Electronic Medical Records (EMRs), since they are in the centre of many current initiatives to rationalize health care services. The American Medical Informatics Association (AMIA) and Health Level Seven (HL7) Inc. have released standard specifications of functions required by such systems. We start by introducing these requirements and a set of typical use case scenarios before discussing architecture and infrastructure for MIS.

We discuss MIS both from a macroscopic perspective as well as from a microscopic perspective. In the latter, we discuss the *architectures* of single systems, their key components and any relevant tools, concepts and techniques to provision and maintain them. In the macroscopic perspective, we view MIS as parts of a large scale health information network that may span health authorities, states, countries or even international jurisdictions.

Here we emphasize the *infrastructure* aspect, which is essential to achieve interoperability and to accommodate asynchronous evolution of network components. We discuss both aspects, architecture and infrastructure, using several concrete case studies from different countries.

3. Health Data and Information

This topic area concentrates on international data representation standards, methodologies and skills needed to work with structured and semi-structured medical data. An overview of HL7 interoperability standards is given, followed by a more detailed introduction of the newest HL7 standard version 3.1 and its model-driven development method [2]. The HL7 Reference Information Model (RIM) will be discussed. We will present a critical assessment of the maturity of this standard and also mention alternatives and related initiatives such as OpenEHR [3]. We survey and put in context several systems for coding health data entries in machine interpretable form, e.g., LOINC for lab results and SNOMEDCT for signs, diseases and interventions.

Two important aspects of implementing health data structures are *repository design* and *messaging*. The first aspect pertains to representing health data in data bases and accessing this data efficiently. The second aspect addresses the data interoperability requirement. While international standards exist only for the second aspect, we discuss both aspects and give examples for their implementation in practice. We also address the issue of data transformation and mediation in order to resolve heterogeneity.

Another important aspect is how to represent medical knowledge in order to support clinical and public health decision making. We will present and contrast alternative computer-based guideline and decision support formalisms [4].

4. Usability engineering

Factors of usability and ergonomics are of key importance for the adoption of MIS solutions in practice. We describe where methods and models of Human Computer Interaction (HCI) have been and could be applied to engineering MIS [5]. We will introduce an HCI framework that parallels the "biopsychosocial" model of care delivery [6]. This discussion will cover the application of HCI evaluation methods to individually focused cognitive models and to more socially focused models

from Computer Supported Collaborative Work (CSCW), e.g., as required for supporting groups of collaborating clinicians.

Also threaded through this discussion will be the positive impact of user centered design approaches to the development of software applications for healthcare. Real world examples at multiple stages (concept, design, development, testing, deployment) will be used to illustrate both positive and negative issues of engagement on the adoption of systems.

5. Privacy & Security

Health information is highly sensitive and must be managed securely to ensure the privacy of individuals. Privacy legislations and standard policies have been introduced in many countries to govern the use of health information, e.g., the Health Insurance Portability and Accountability Act (HIPAA). We discuss the impact of these legislations and standards on the MIS requirements development. We survey ways to elicit and capture multilateral security and privacy requirements and present a case study [7]. We discuss the issue of managing patient consent and principle patterns for implementing typical security and privacy requirements. We also address the tension between security and usability requirements and refer to experiences with good and bad practices as documented in literature.

6. References

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